

MINI PROJECT REPORT

COVID-19 Exploratory Data Analysis

India scope validation and early-outbreak snapshot analysis

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Dataset	2019_nCov_20200121_20200126 - SUMMARY.csv
Prepared on	05 September 2026

Important scope statement. The required CSV has no country row labelled India. This report therefore records the data-validation result and analyses only the global / source-label evidence present in the supplied file. It does not create, infer, or claim India-specific statistics.

Prepared as an original exploratory analysis using the provided dataset and reproducible Python workflow.

1. Executive summary

The supplied historical CSV contains **368 records**, **7 fields**, and reports dated from **21 January 2020 to 26 January 2020**. It contains **21 source country labels** but **zero India-labelled records**. The assignment title is therefore retained for submission continuity, while the analysis scope is corrected to the evidence actually available.

Using a documented representative-report rule, the last selected report is 26 January 2020, 11:00 AM. It contains 2,116 reported confirmed cases, 383 suspected values, 52 recovered values, and 56 deaths. Reported confirmed values increase from 332 to 2,116 over the selected dates (537.3% increase). These numbers are descriptive values from the file, not an adjusted or official epidemiological estimate.

Indicator	Result	Interpretation
Raw records	368	All rows in supplied CSV
India-labelled records	0	India-specific EDA is not possible with this file
Selected report dates	6	One representative report per calendar day
Final report timestamp	26 Jan 2020, 11:00 AM	Latest selected timestamp
Final reported confirmed	2,116	Sum of recorded numeric values
Final reported deaths	56 (2.65%)	Descriptive ratio to recorded confirmed values

2. Data understanding and quality checks

The original fields are Province/State, Country, Date last updated, Confirmed, Suspected, Recovered, and Deaths. The timestamp column mixes date-only strings with date-time strings, so it is parsed with a mixed-format method and a version-compatible fallback. Blank metric cells are kept visible in the audit; numerical totals sum recorded values only and should not be read as proof that a blank field equals zero.

Field	Blank cells	Missing share
Deaths	344	93.5%
Recovered	332	90.2%
Suspected	280	76.1%
Province/State	64	17.4%
Confirmed	29	7.9%
Date last updated	0	0.0%
Country	0	0.0%

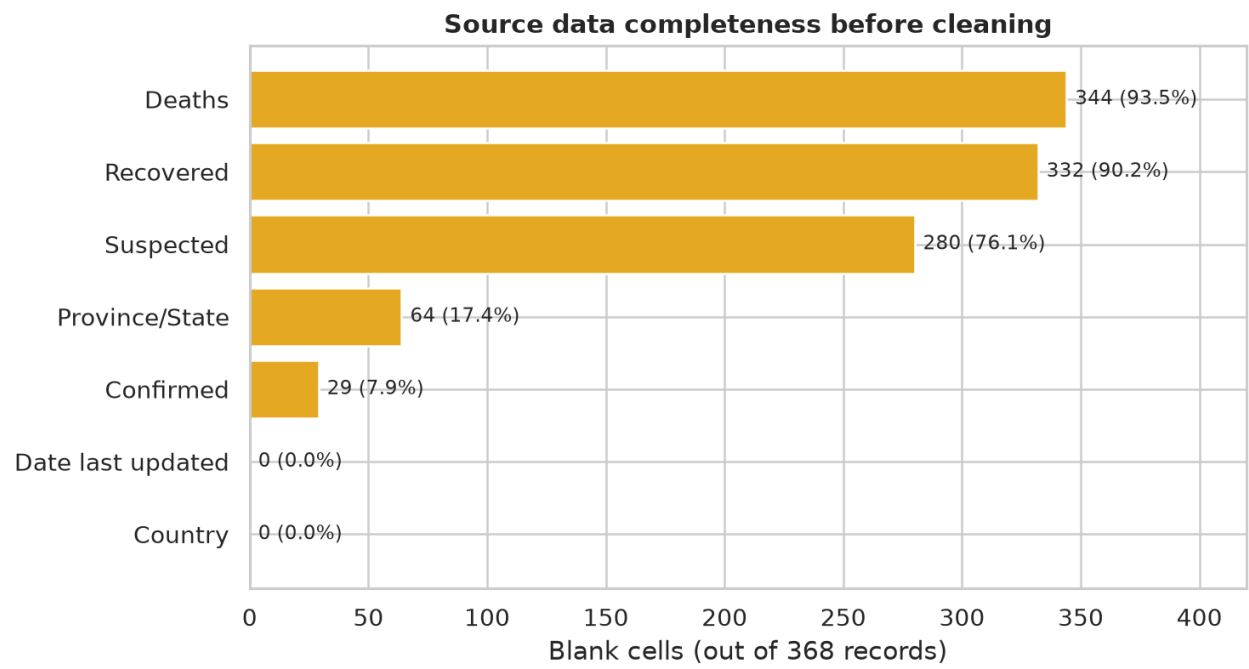


Figure 1. Source completeness prior to cleaning. The health-outcome fields contain many blanks, which limits the interpretation of rates.

3. Representative-report method

Several calendar days have more than one timestamp. A direct sum across all such rows would count the same location more than once or combine a partial supplement with a fuller report. For each calendar date, this analysis: (1) calculates the number of rows and distinct country-province location keys at each timestamp; (2) excludes a timestamp where a location repeats; and (3) selects the clean timestamp with the greatest distinct-location coverage, using the later timestamp only to break a tie. This rule is transparent and reproducible, but it does not turn the dataset into an official surveillance series.

Date	Selected report time	Rows	Distinct locations	Reported confirmed
21 Jan	12:00 AM	27	27	332
22 Jan	12:00 PM	38	38	555
23 Jan	12:00 PM	46	46	653
24 Jan	12:00 PM	40	40	939
25 Jan	12:00 AM	44	44	1,354
26 Jan	11:00 AM	46	46	2,116

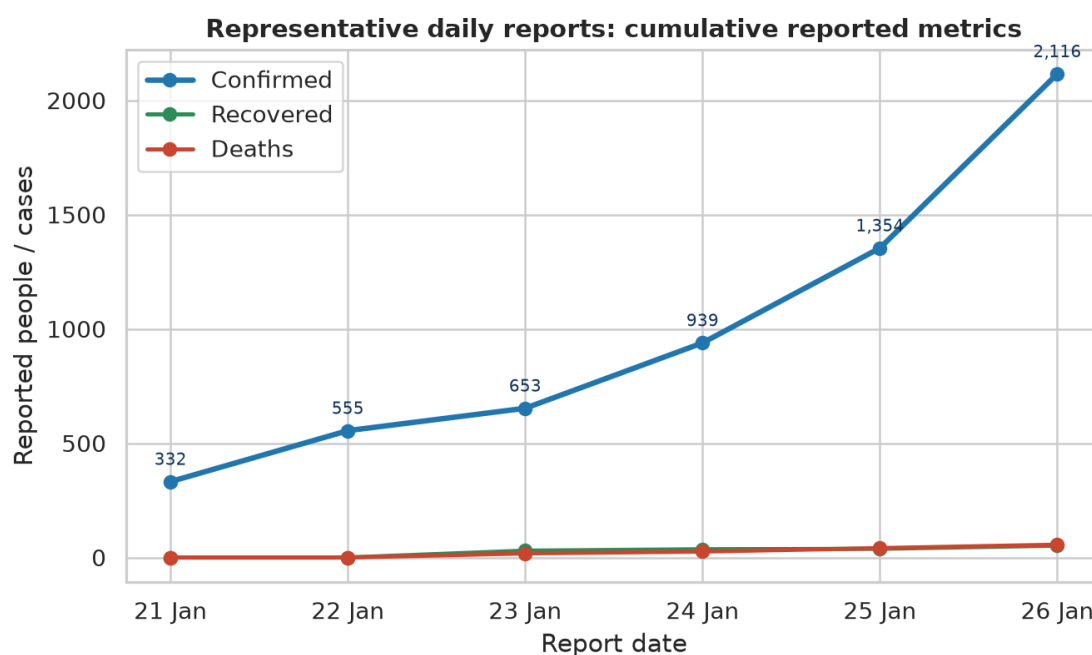


Figure 2. Cumulative values in the selected representative reports. Different report times and incomplete fields make this a descriptive trend only.

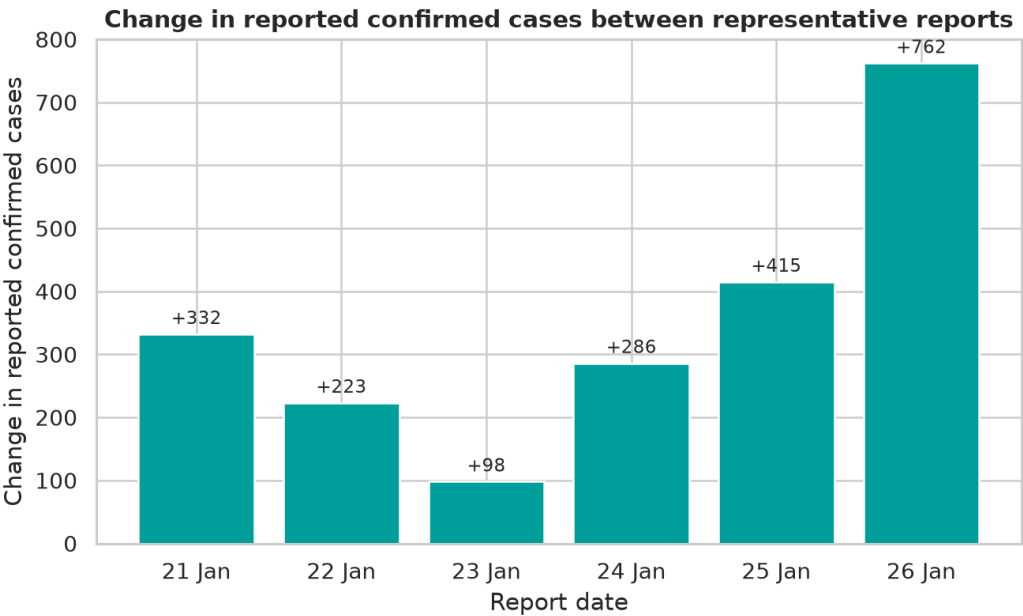


Figure 3. Change in the reported confirmed value between consecutive representative reports.

4. Final-report exploration

The final selected report contains 2,116 recorded confirmed values. The source label **Mainland China** accounts for 2,062 (97.45%) of those values; the remaining source labels together account for 54. Source labels are displayed as provided (apart from the US/United States spelling standardisation), rather than being reclassified by this report.

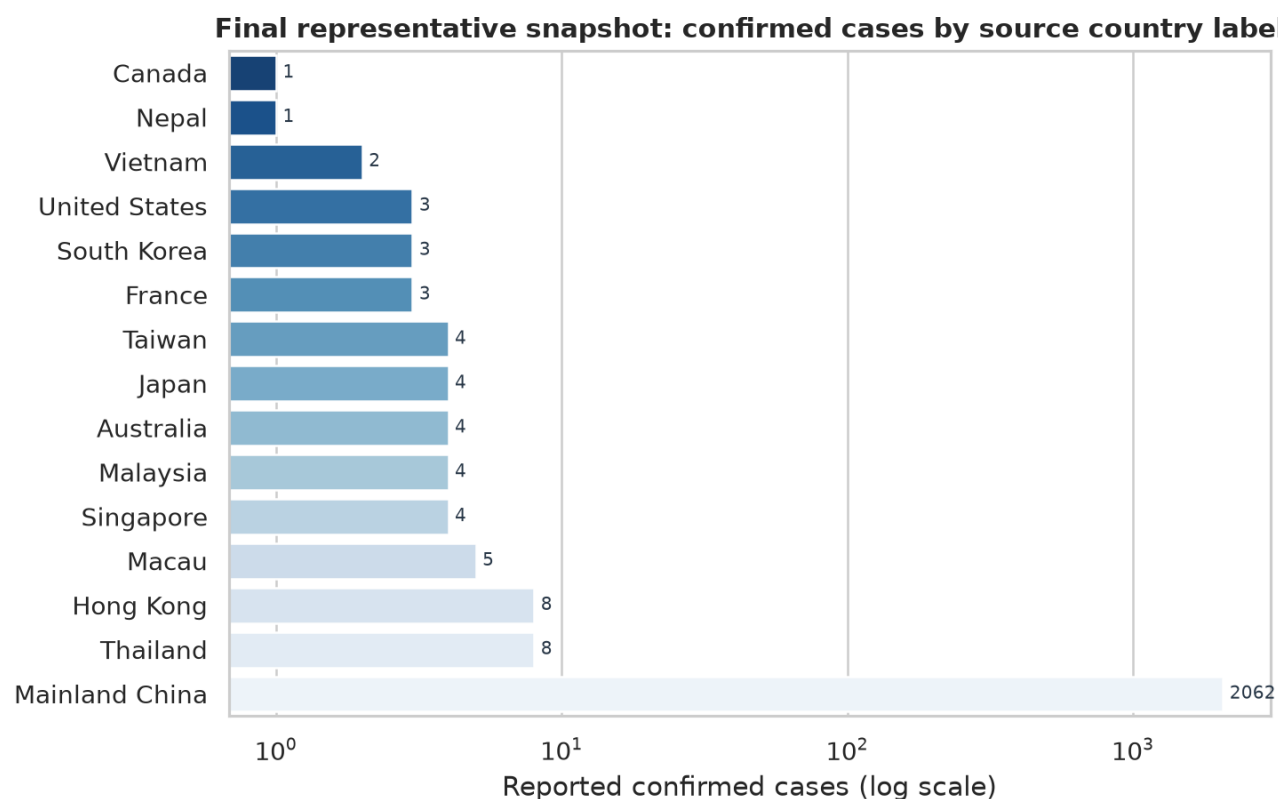


Figure 4. Final representative report by source country label. A log scale preserves visibility for the smaller values.

Source country label	Confirmed	Share of final total
Mainland China	2,062	97.45%
Hong Kong	8	0.38%
Thailand	8	0.38%
Macau	5	0.24%
Australia	4	0.19%
Malaysia	4	0.19%

5. Provincial concentration in the source's Mainland China label

Within rows labelled Mainland China, Hubei records 1,058 confirmed values. That is 51.31% of the 2,062 values in this source-label group and 50.00% of the entire final selected total. This concentration is descriptive of the supplied early-outbreak report and should not be generalized to a later period.

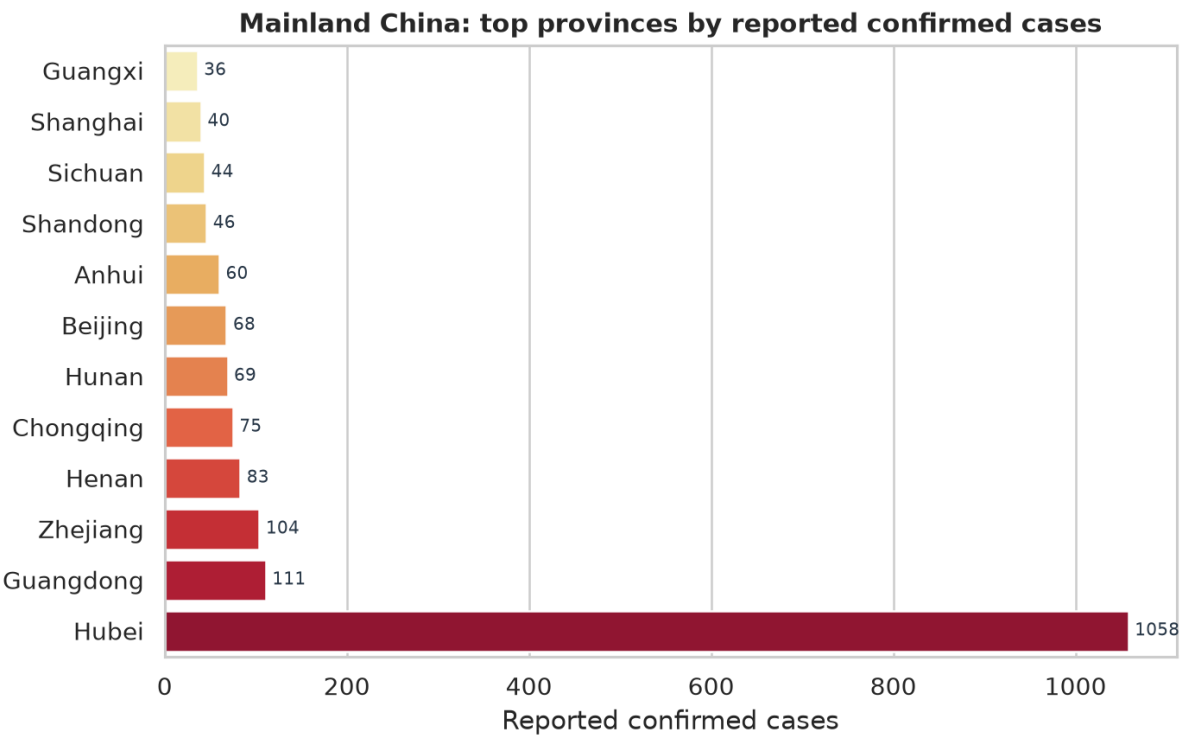


Figure 5. The twelve provinces with the largest recorded confirmed values in the final representative report.

Concentration of reported cases within Mainland China

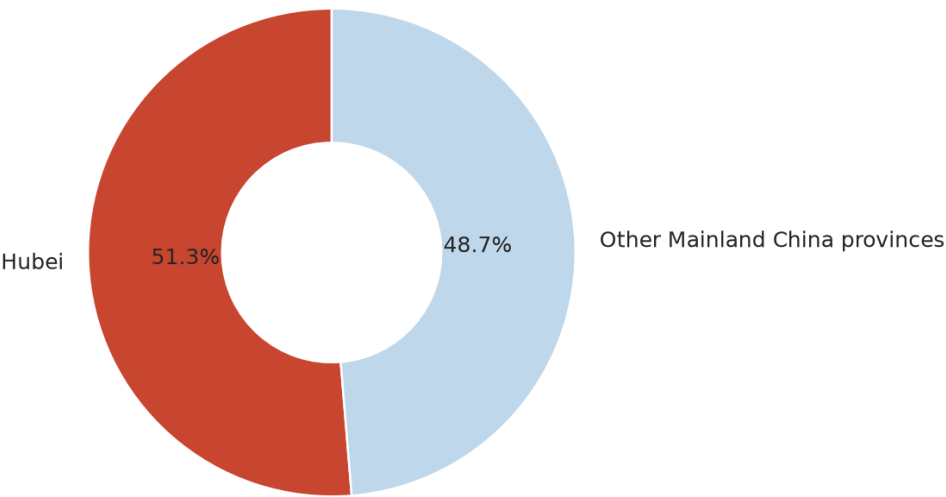


Figure 6. Hubei's share within the source's Mainland China label group.

6. Relationships and interpretation

The next plots are exploratory rather than predictive. Reported suspected values are frequently blank, and the source-label observations are few and highly concentrated. The scatter plot is useful for seeing the documented distribution, while the correlation heatmap only describes co-movement within this small final snapshot. Neither chart establishes a causal relationship.

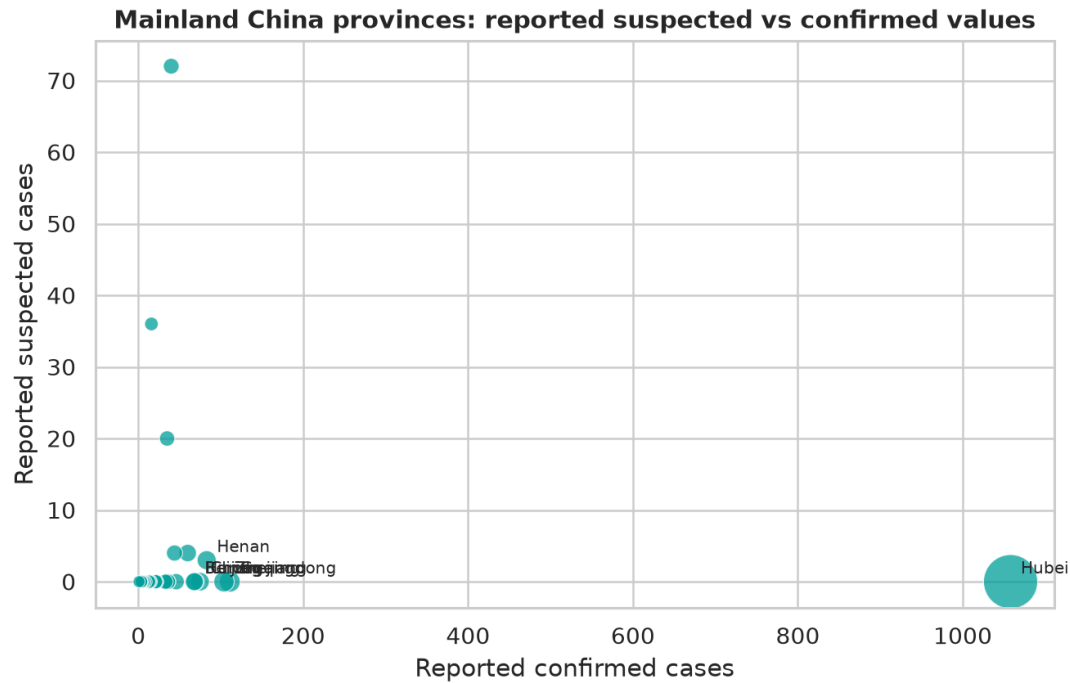


Figure 7. Province-level relationship between reported suspected and confirmed values in the final report.

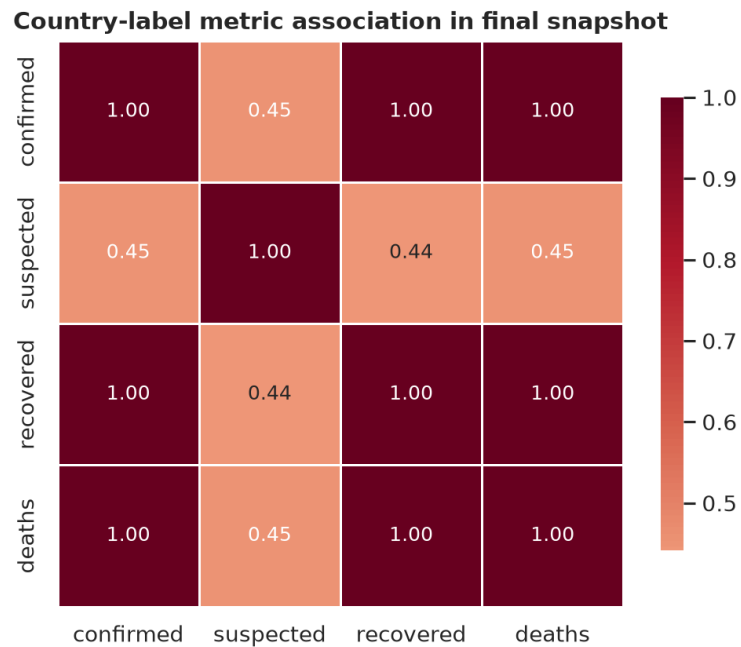


Figure 8. Descriptive correlation between recorded country-label metrics in the final representative report.

7. Conclusions and limitations

1. The requested CSV is a small early-outbreak dataset, not an India state-wise COVID-19 dataset. Its most important India-related finding is that India is absent.
2. After avoiding repeat-location records, the chosen representative reports rise from 332 to 2,116 recorded confirmed values over 21-26 January 2020.
3. The final selected report is highly concentrated: Hubei alone contains 1,058 recorded confirmed values, and Mainland China is the dominant source label.
4. Heavy missingness in suspected, recovered, and death fields prevents robust rate comparisons; no missing values were imputed.
5. No claim, recommendation, or model is made for India from these data. A valid India-specific extension would require an India-labelled, state-level dataset covering the relevant period.

Reproducibility

Files submitted: `2019\_nCoV\_20200121\_20200126 - SUMMARY.csv`, `MiniProject\_COVID19\_India\_EDA.ipynb`, and this PDF. The notebook reads the local CSV and uses Python packages pandas, NumPy, Matplotlib, and Seaborn. It includes all cleaning, snapshot selection, tables, and figures used in this report.

Data provenance

Dataset filename supplied for the task: **2019_nCoV_20200121_20200126 - SUMMARY.csv**. The analysis uses only that file. The notebook and report are original work prepared for this submission; they do not substitute data from a different India source.

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